

AKSHAYA BHARADHWAJ

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SUMMARY

Data science graduate student (UCI MDS, GPA 3.93) with experience applying machine learning, statistical evaluation, and Python-based data workflows to healthcare imaging, manufacturing, and operational data. Skilled in model development, experimental evaluation, data preprocessing, and translating analytical results into actionable insights.

EDUCATION

University of California, Irvine, CA

Master of Data Science, GPA: 3.93/4.00

December 2026

Relevant Coursework: Intermediate Probability and Statistics I, Introduction to AI, Databases and Data Management

Mukesh Patel School Of Technology, Management & Engineering, SVKM's NMIMS, Mumbai, India

Bachelor of Technology, Data Science, GPA: 3.74/4.00

August 2025

Relevant Coursework: Statistics, Data Visualization and Analysis, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

TECHNICAL SKILLS

Languages: Python, SQL, R

Machine Learning: Classification, Regression, Deep Learning, Computer Vision, NLP, Model Evaluation, Error Analysis

Statistics & Experimentation: Cross-Validation, Hypothesis Testing, Statistical Testing, Experimental Design

Data: Data Cleaning, Validation, Transformation, MySQL

Libraries: scikit-learn, Pandas, NumPy, PyTorch, TensorFlow

PROFESSIONAL EXPERIENCE

Data Science Intern, Pediatric Imaging - Children's Hospital of Orange County (CHOC), Orange, CA

July 2026 – September 2026

- Developed and evaluated a custom lightweight CNN for 3-class necrotizing enterocolitis (NEC) classification from neonatal abdominal X-rays.
- Conducted controlled architecture and augmentation experiments, systematically varying convolutional depth and evaluating geometric, pixel-level, and physics-informed noise transformations across multiple random seeds.
- Built a reproducible experimental pipeline with patient-level data splitting and held-out test-set isolation, using Macro-F1 and Macro ROC-AUC for model selection and evaluation.

Graduate Research Assistant - Dept. of Statistics, UC Irvine, CA

July 2026 – Present

- Building a benchmark dataset linking LoL replay video frames with exported time-series stats (economy/map control/tech progression) to study "turning points" in match outcomes
- Prototyping turning-point detection approaches using change-point detection + vision-language models (VLMs) and comparing signals against existing win-probability baselines

WIPRO LIMITED

Research Intern - Machine learning at Wipro Infrastructure Engineering (Wipro PARI)

December 2024 – August 2025

- Built and evaluated ANN models using Python and scikit-learn, validated using R^2 , to model complex Li-ion battery manufacturing processes, enabling analysis of how upstream parameters impact downstream material quality and reliability.
- Developed a process-parameter framework spanning 16 manufacturing stages in collaboration with industry experts, supporting data-driven process optimization across the EV battery pipeline.
- Contributed to 2 review papers and a research paper proposing a machine-learning-driven framework for EV battery process optimization, currently under submission to academic and industry venues.

Data Science Intern - Wipro Water, Navi Mumbai, India

May 2024 – July 2024

- Cleaned and analyzed multi-city operational datasets to uncover trends supporting business decision-making.
- Built Power BI visualizations and presented findings to the marketing team and CEO using clear, non-technical explanations.
- Automated Python workflows to validate and format raw data for Salesforce lead creation, improving data quality and operational efficiency.

Intern, IDEA CREST SOLUTIONS PRIVATE LTD, Bangalore, India

May 2023 – July 2023

- Developed RESTful APIs using Python (Flask) and optimized MySQL database schemas for a streaming platform management system, improving data integrity and query performance.
- Created Postman collections to test endpoints and document API workflows, improving handoffs and reducing debugging time for integrations.

RESEARCH & PROJECT EXPERIENCE

Automated Web Data Extraction via Zero-Shot NLP Classification | [Published](#) | ICDAM-2025

Nov 2024 – May 2025

- Built a data extraction system using Zero-Shot Learning and NLP classification; published in Elsevier SSRN and presented at ICDAM-2025.

ValidClean: Multi-Agent Agentic AI Framework for Data Preprocessing - Presented at UCI GradSLAM 2026

February 2026 – Present

- Architected a 5-agent LLM system to profile datasets, infer ML objectives, and generate preprocessing strategies; validated impact using K-fold cross-validation, paired t-tests, and Cohen's d.

Raga Recognition from Audio: End-to-End ML Classification | DATA 210P Statistical Methods I Course Project

March 2026

- Engineered a 228-dimensional hybrid feature vector and trained 5 classifiers (SVM, RF, XGBoost, MLP, Ensemble) on the Saraga Carnatic dataset, achieving 77.8% accuracy (XGBoost) vs 33.3% random baseline under severe data scarcity using GroupKFold cross-validation.

Skin Cancer Detection from Dermoscopic Images | ML Course Project | COMPSCI 273P

March 2026

- Developed a multimodal CNN + MLP pipeline on 400k+ dermoscopic images (ISIC 2024), achieving 0.93 recall and 0.942 ROC-AUC under 0.1% class imbalance.